Event-TimeRAF: Event-Aware Retrieval-Augmented Foundation Model for Explainable Air Quality Forecasting Under Concept Drift

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Abstract—**Short-horizon PM2.5 forecasting is complicated by meteorological variation, calendar effects, environmental events, and distribution shift. Although time-series foundation models provide transferable forecasting capability and TimeRAF shows how historical retrieval can augment a frozen forecaster, numerically similar windows need not share the local conditions that govern urban air quality. This study presents Event-TimeRAF, an event-aware retrieval framework comprising a Source-Audited Context Builder, a Leakage-Safe Event-Context Retriever, and a Drift-Aware Forecast and Evidence Head. The framework aligns official air-quality, weather, calendar, and storm-event records; restricts retrieval to training-history windows whose targets precede the query lookback; evaluates feature-level XGBoost and forecast-level frozen Chronos-Bolt fusion; and stores traceable explanation evidence. Evaluation uses EPA AirData PM2.5, NOAA Global Hourly weather, and hash-verified NOAA Storm Events records for Los Angeles County from 2019 through 2024, with a 168-hour lookback and a 24-hour horizon. The strongest pre-specified supervised context baseline obtains MSE 26.185, MAE 3.125, RMSE 5.117, and R^2 0.379, whereas the full feature-level Event-TimeRAF model obtains MSE 26.712. Frozen Chronos-Bolt obtains MSE 28.941, and retrieval fusion reduces it to 28.709, although the paired confidence interval crosses zero. These values are recomputed from 7,199 test origins and manifest-backed prediction artifacts. The evidence supports selective rather than uniform retrieval gains. Because Storm Events lacks machine-readable publication timestamps, event-conditioned findings are retrospective sensitivity results, not strict real-time forecasts.**

Index Terms—**Air quality forecasting, PM2.5, retrieval-augmented forecasting, time-series foundation models, TimeRAF, concept drift, explainable forecasting.**

I. Introduction

Fine particulate matter (PM2.5) is a major urban air-quality risk because it is associated with respiratory and cardiovascular health effects [1]. Accurate short-term PM2.5 forecasts can support public-health alerts, exposure reduction, transport planning, and environmental operations. In Los Angeles County, however, PM2.5 dynamics are difficult to model with pollutant history alone. Concentrations can change with wind speed, boundary-layer behavior, rainfall, seasonal activity, holidays, wildfire smoke, and unusual storm or high-wind events. These factors make the forecasting problem both temporal and contextual.

Recent time-series foundation models (TSFMs) aim to improve forecasting generalization by pre-training on large time-series corpora and transferring the learned forecasting ability to new domains. Representative models include TimesFM, Moirai, MOMENT, Chronos, and Tiny Time Mixers [2]–[6]. In parallel, retrieval-augmented generation has shown that non-parametric memory can help foundation models access relevant external knowledge at inference time [7], [8]. TimeRAF combines these two ideas for zero-shot time-series forecasting by retrieving relevant historical sequences from a curated knowledge base and integrating them into a frozen TSFM through Channel Prompting [9].

Air-quality forecasting raises a domain-specific retrieval question that is not fully addressed by numerical time-series retrieval alone. A historical PM2.5 window may look similar to the current window but occur under different weather or event conditions. Conversely, a window with moderate numerical similarity may be more informative if it shares high-wind, rainfall, holiday, smoke, or stagnation context. This motivates a retrieval framework that treats source records, weather, calendar information, and distribution-shift indicators as part of the forecasting evidence rather than as unrelated metadata.

This study introduces Event-TimeRAF, an event-aware retrieval-augmented framework for explainable PM2.5 forecasting under distribution shift. The method adapts the retrieval principle of TimeRAF to an official-source environmental setting without claiming to reproduce its learned retriever or Channel Prompting mechanism. Event-TimeRAF is organized into three named components: a Source-Audited Context Builder (SACB), a Leakage-Safe Event-Context Retriever (LSER), and a Drift-Aware Forecast and Evidence Head (DFEH). Together, these components address three questions: how official environmental records can be converted into a retrieval-compatible context; how historical candidates can be selected without temporal leakage; and when retrieval contributes beyond a strong supervised weather-calendar baseline.

The main contributions are as follows:

- *Source-Audited Context Builder:* a reproducible mechanism that aligns EPA PM2.5, NOAA weather, calendar, and hash-verified NOAA Storm Events records while preserving source identity, coverage checks, and the declared event-availability assumption.
- *Leakage-Safe Event-Context Retriever:* a historical analogue mechanism that combines PM2.5 shape, weather, calendar, and event similarity after enforcing an embargo under which every candidate target ends before the query lookback begins.
- *Drift-Aware Forecast and Evidence Head:* an evaluation layer that exposes direct XGBoost forecasts, frozen Chronos-Bolt retrieval fusion, transparent shift indicators,

and machine-readable explanation evidence under one chronological protocol.

The evaluation is deliberately conservative. It restricts reported model comparisons to the manifest-backed run, reports uncertainty for paired comparisons, and does not interpret event association as causation. The findings therefore characterize where retrieval is useful and where additional evidence is required before operational event-aware forecasting can be claimed.

II. Literature Review and Related Work

A. Statistical, Machine-Learning, and Deep Forecasters

Forecasting methods differ in how they trade inductive bias, data demand, and computational cost. Autoregressive and seasonal models remain informative baselines because their failure modes are easy to inspect. Tree ensembles such as XGBoost and LightGBM extend this setting to nonlinear lag, rolling, weather, and calendar interactions without requiring a sequence encoder [10], [11]. Recurrent LSTM and GRU models instead learn stateful temporal representations [12], [13]. Large benchmark archives, including the Monash Forecasting Repository, further support reproducible comparison across heterogeneous time-series domains [14]. Transformer families emphasize longer dependencies: Informer reduces attention cost, Autoformer and FEDformer use decomposition, Crossformer and iTransformer model cross-variable structure, and TimesNet reorganizes temporal variation into two-dimensional patterns [15]–[20]. These gains do not imply that greater architectural complexity is always necessary. DLinear and PatchTST show that simple decomposition or patch-based inductive biases can remain highly competitive [21], [22], whereas TimeMixer uses explicit multiscale decomposition and mixing [23]. Event-TimeRAF therefore retains seasonal and boosted tree baselines instead of treating a deep backbone as the default winner.

B. Time-Series Foundation Models

Time-series foundation models (TSFMs) seek transfer across datasets rather than training a separate representation from scratch for each series. TimesFM uses a decoder-only forecasting design; Moirai trains a universal forecasting Transformer; MOMENT supports several time-series tasks; Chronos represents values as tokens; and Tiny Time Mixers emphasizes compact zero- and few-shot transfer [2]–[6]. Timer, Timer-XL, UniTime, and Time-LLM explore generative pretraining, long-context scaling, cross-domain unification, or language-model reprogramming [24]–[27]. Frozen pretrained models have also been evaluated as general computation engines for time-series analysis [28]. Collectively, these studies motivate zero-shot evaluation, but they also show that transfer performance depends on input representation and domain shift. The present study therefore compares frozen Chronos-Bolt against locally supervised models on identical forecast origins rather than assuming foundation-model superiority.

C. Retrieval-Augmented Forecasting

Retrieval augmentation separates parametric knowledge from an external memory. In natural-language processing, RAG and Dense Passage Retrieval established the retrieve-then-condition pattern [7], [8]. In time series, retrieval has been studied through diffusion forecasting and assisted generation workflows [29], [30]. TimeRAF provides the closest methodological reference: it constructs a time-series knowledge base, trains an MLP dual-encoder retriever, selects top-k candidates, and integrates candidate embeddings into a frozen TSFM through Channel Prompting [9]. Its ablations separate retrieval quality, integration, candidate count, knowledge-base composition, and backbone choice.

Event-TimeRAF retains the external-memory principle but addresses a different experimental question. It uses transparent random, cosine, and event-context similarity instead of a learned dual encoder, and it uses feature- or forecast-level fusion instead of internal Channel Prompting. This distinction prevents the current implementation from being described as a reproduction of TimeRAF; it is an audited environmental adaptation designed to test whether contextual historical analogues help a local PM2.5 task.

D. Air-Quality Forecasting and Official Context

PM2.5 forecasting commonly combines pollutant history with temperature, humidity, wind, precipitation, and pressure. Recent reviews describe a shift from single-station recurrent models toward attention-based, hybrid, and spatiotemporal systems [31], [32]. AirFormer, for example, separates deterministic spatiotemporal modeling from stochastic uncertainty estimation at nationwide scale [33]. Such multi-station designs address spatial dependence, whereas the present study examines a county-level aggregate and the provenance of contextual evidence. EPA AirData supplies regulated PM2.5 observations, NOAA Integrated Surface Database supplies hourly meteorology, and NOAA Storm Events supplies structured hazard records [34]–[36]. NOAA Hazard Mapping System products could add smoke evidence but were not included in the reported run [37].

E. Distribution Shift and Forecast Explanations

Temporal nonstationarity can alter means, variances, and relationships between inputs and future values. Adaptive normalization methods address this problem by estimating statistics at local temporal scales [38]; Event-TimeRAF instead uses interpretable shift indicators to audit regimes without claiming that shift has been eliminated. Explanation methods also differ in scope. SHAP attributes predictions to model inputs [39], while information-theoretic saliency supports counterfactual inspection of probabilistic forecasts [40]. The implemented explanation layer is narrower: it records signed XGBoost effects, retrieved cases, event identifiers, drift components, and an uncertainty proxy. These records support traceability, but they do not establish causal effects of weather or events.

III. Critical Gaps and Limitations of Previous Studies

Prior work leaves three connected gaps for event-sensitive urban air-quality forecasting. First, temporal forecasters and numerical retrieval systems do not necessarily preserve the provenance, coverage, and availability assumptions of local contextual records. A numerically similar PM2.5 window can occur under a different weather or event regime, and a retrospective event label can be mistaken for information available at forecast time. The SACB addresses this gap by joining official records under explicit source, coverage, and availability audits.

Second, historical retrieval can introduce look-ahead bias when a candidate input or target overlaps the query context. Similarity alone does not prevent a near-duplicate future from entering the retrieved evidence. The LSER addresses this gap through a strict temporal embargo and training-history restriction, followed by separately auditable PM2.5, weather, calendar, and event similarity terms.

Third, retrieval gains are often reported without testing whether a strong local context model already captures the useful signal, whether the gain persists under distribution shift, or whether an explanation can be traced to stored evidence. The DFEH addresses this gap by evaluating direct XGBoost and frozen Chronos-Bolt paths on common origins, exposing shift indicators, and retaining machine-readable feature, retrieval, event, and uncertainty evidence. This component supports comparison and auditability; it does not convert contextual association into causal explanation.

These mappings delimit the novelty of the study. Event-TimeRAF does not contain a learned dual-encoder retriever or internal Channel Prompting, and the present single-county experiment does not establish geographic generalization.

IV. Core Contributions

The proposed framework contributes three mechanisms corresponding one-to-one with the gaps above:

- *Source-Audited Context Builder (SACB):* constructs an aligned 85-dimensional context from official PM2.5, meteorological, calendar, and event records while retaining source and availability metadata.
- *Leakage-Safe Event-Context Retriever (LSER):* retrieves training-history analogues only after enforcing a candidate-target embargo, then exposes the contribution of each similarity channel and the retrieved trajectory evidence.
- *Drift-Aware Forecast and Evidence Head (DFEH):* places direct horizon models, frozen-TSFM forecast fusion, drift-state analysis, and traceable explanation records within one chronological evaluation contract.

The empirical contribution is intentionally qualified: weather-calendar context produces the clearest supported gain, whereas event and retrieval components provide selective rather than uniformly positive improvements.

TABLE I
Main Symbols Used in the Proposed Formulation

Symbol	Meaning
L	Lookback window length
H	Forecast horizon
X_t	Historical PM2.5 sequence at time t
W_t	Weather covariates
C_t	Calendar/context features
E_t	Event context available at the forecast origin
R_t^{ts}	Retrieved time-series knowledge
R_t^{ev}	Retrieved event knowledge
D_t	Drift indicators
q_t	SACB context vector in $\mathbb{R}^{85}$
G_t	LSER top-k retrieved set
$\rho(G_t)$	Retrieval summary in $\mathbb{R}^{51}$
$\mathcal{O}_t$	Machine-readable DFEH evidence record
$\hat{Y}_{t+1:t+H}$	Predicted future PM2.5 values

V. Problem Formulation

Let x_t denote the PM2.5 measurement at time t. For a forecast time t, the historical air-quality lookback window is

$$X_t = [x_{t-L+1}, x_{t-L+2}, \ldots, x_t], \tag{1}$$

where L is the lookback length. Let W_t denote weather covariates aligned with the lookback window, C_t denote calendar features, and E_t denote event context available under the recorded source assumption at the forecast origin. The target associated with the input in (1) is

$$Y_{t+1:t+H} = [x_{t+1}, x_{t+2}, \ldots, x_{t+H}], \tag{2}$$

where H is the forecasting horizon. Equation (2) therefore defines the 24-step vector evaluated at every valid origin.

The forecasting goal is to learn a function

$$\hat{Y}_{t+1:t+H} = F_\theta(X_t, W_t, C_t, E_t, R_t^{ts}, R_t^{ev}, D_t), \tag{3}$$

where R_t^{ts} is retrieved time-series knowledge, R_t^{ev} is retrieved event knowledge, and D_t contains concept-drift indicators. The implemented target setting is $L = 168$ hours and $H = 24$ hours. Equation (3) states the full information interface; the implemented modules that construct these arguments are specified in the next section.

The evaluation uses a time-based split:

$$\mathcal{T}_{train} < \mathcal{T}_{val} < \mathcal{T}_{test}, \tag{4}$$

where the earliest 70% of observations are used for training, the next 15% for validation, and the latest 15% for testing. The ordering in (4) is retained for feature fitting, retrieval eligibility, model selection, and final evaluation.

Table I summarizes the essential notation.

VI. Methodology

Event-TimeRAF comprises three top-level components. The Source-Audited Context Builder (SACB) converts heterogeneous official records into a chronological hourly context while retaining provenance and availability metadata. The Leakage-Safe Event-Context Retriever (LSER) constructs an embargoed historical memory and selects analogues according to PM2.5, weather, calendar, and event similarity. The Drift-Aware Forecast and Evidence Head (DFEH) exposes feature-level and forecast-level prediction paths, shift indicators, and a machine-readable explanation record. These components are deterministic or supervised modules over one-dimensional time-series and tabular inputs; no image operator, learned attention block, or internal TimeRAF Channel Prompting is implied.

A. Framework Overview and Forward Pass

Figure 1 summarizes the complete dataflow. EPA PM2.5, NOAA weather, calendar, and NOAA Storm Events records first enter the SACB, which audits, aligns, and transforms them into a query context and input–target windows. The LSER filters the training-history knowledge base, ranks eligible candidates, and returns aligned candidate futures and retrieval statistics. The DFEH then supplies those statistics to 24 direct XGBoost regressors, evaluates a separate frozen Chronos-Bolt fusion path, computes drift evidence, and writes the forecast and its supporting record.

For each forecast origin t, Event-TimeRAF maps the aligned context into a forecast through

$$\hat{Y}_{t+1:t+H} = F_\theta\big(\phi(X_t, W_t, C_t, E_t), G_t, D_t\big), \tag{5}$$

where $\phi(\cdot)$ denotes the tabular feature construction, G_t denotes retrieved historical evidence, and D_t denotes drift evidence. The key methodological constraint is that every element of G_t and D_t must be available no later than the forecast origin. Equation (5) is the compact functional form that the tensor-level forward pass below expands.

For a batch of B origins, the implemented forward pass can be expanded in tensor notation as

$$\begin{aligned}
Q &= \mathrm{SACB}(X, W, C, E) && \in \mathbb{R}^{B\times 85},\\
G &= \mathrm{LSER}(Q, \mathcal{K}) && \in \mathbb{R}^{B\times 8\times 24},\\
U_h &= [Q \,\|\, \rho(G) \,\|\, d \,\|\, C_h^+] && \in \mathbb{R}^{B\times 151},\\
\hat{Y}^X &= [f_1(U_1), \ldots, f_{24}(U_{24})] && \in \mathbb{R}^{B\times 24},\\
(\hat{Y}, \mathcal{O}) &= \mathrm{DFEH}(\hat{Y}^X, \hat{Y}^C, \hat{Y}^R, d, G) && \in \mathbb{R}^{B\times 24}\times \mathcal{E},
\end{aligned} \tag{6}$$

where Q is the 85-dimensional origin context, $\rho(G) \in \mathbb{R}^{B\times 51}$ contains the retrieved trajectory, spread, two similarity summaries, and candidate count, $d \in \mathbb{R}^{B\times 6}$ contains five drift components and their mean score, and $C_h^+ \in \mathbb{R}^{B\times 9}$ is the known calendar vector for horizon h. The symbol $\mathcal{E}$ denotes the schema of the evidence record $\mathcal{O}$. Equation (6) describes both reported forecasting paths without presenting the XGBoost path as part of the frozen foundation model.

B. Source-Audited Context Builder

The SACB is motivated by the need to preserve data provenance and temporal availability before model fitting. As shown in Figure 2, source-specific checks precede feature construction. EPA monitor records are aggregated to an hourly county median, NOAA weather is aligned from the selected surface station, event archives are hash-verified against a manifest, and calendar values are derived from the local timestamp. The resulting context is then windowed and split chronologically; target values are never interpolated.

1) Dataset Construction: Hourly PM2.5 observations are taken from US EPA AirData/AQS parameter 88101 for Los Angeles County [34]. The data audit selects one monitoring site using coverage and continuity criteria when a single site satisfies the coverage criterion. Because no individual monitor met this criterion, the target is the hourly county median across official Los Angeles County AQS monitors. Timestamps are converted to `America/Los_Angeles` with explicit daylight-saving handling. Short input gaps may be filled only from past-available observations; targets are never interpolated. Weather covariates come from the nearest suitable NOAA Integrated Surface Database station [35]. Calendar features include cyclic hour, weekday, and month terms, weekend status, US federal holidays, and California holidays.

At each origin, the feature groups are concatenated as

$$q_t = [p_t \,\|\, w_t \,\|\, c_t \,\|\, e_t] \in \mathbb{R}^{23+14+9+39} = \mathbb{R}^{85}, \tag{7}$$

where p_t, w_t, c_t, and e_t are respectively the PM2.5, weather, calendar, and event feature vectors. The PM2.5 group contains fixed lags, rolling moments and extrema, differences, the current value, and a fill flag. The weather group contains six measured variables, a precipitation-missing flag, derived condition flags, a stagnation proxy, and 24-hour rolling means. The event group contains one-hour and active counts, 24/72/168-hour counts, category-specific 72-hour counts, and an event-burst ratio. Thus, Equation (7) fixes both the feature ordering and the 85-dimensional SACB output used by downstream modules.

All normalization statistics are fitted on the training split only. Query and candidate windows use identical normalization, following the TimeRAF principle that retrieved and input sequences must share a preprocessing convention. The reported run contains 48,332 valid windows with $X \in \mathbb{R}^{48332\times 168}$ and $Y \in \mathbb{R}^{48332\times 24}$.

C. Leakage-Safe Event-Context Retriever

The LSER separates temporal eligibility from relevance. Figure 3 first shows the embargo: a candidate is considered only if its complete target precedes the beginning of the query lookback. The eligible candidates are then compared along four independently stored similarity channels. The selected futures are aligned to the query scale and summarized for the forecasting head. This ordering prevents a high similarity score from overriding the temporal constraint.

1) Time-Series Knowledge Base: The time-series knowledge base contains historical PM2.5 windows. Each record stores a window identifier, start time, end time, input and future values, weather summary, calendar summary, normalized window vector, split, and normalization identifier. Knowledge-base origins are separated by 192 hours, equal to the 168-hour input and 24-hour future combined. The resulting reported knowledge base contains 191 non-overlapping training windows.

Formally, the knowledge base is

$$\mathcal{K} = \{(u_i, X_i, Y_i, m_i, c_i, e_i)\}_{i=1}^{N}, \tag{8}$$

where u_i is the candidate origin, X_i and Y_i are the candidate lookback and future windows, m_i is a weather summary, c_i

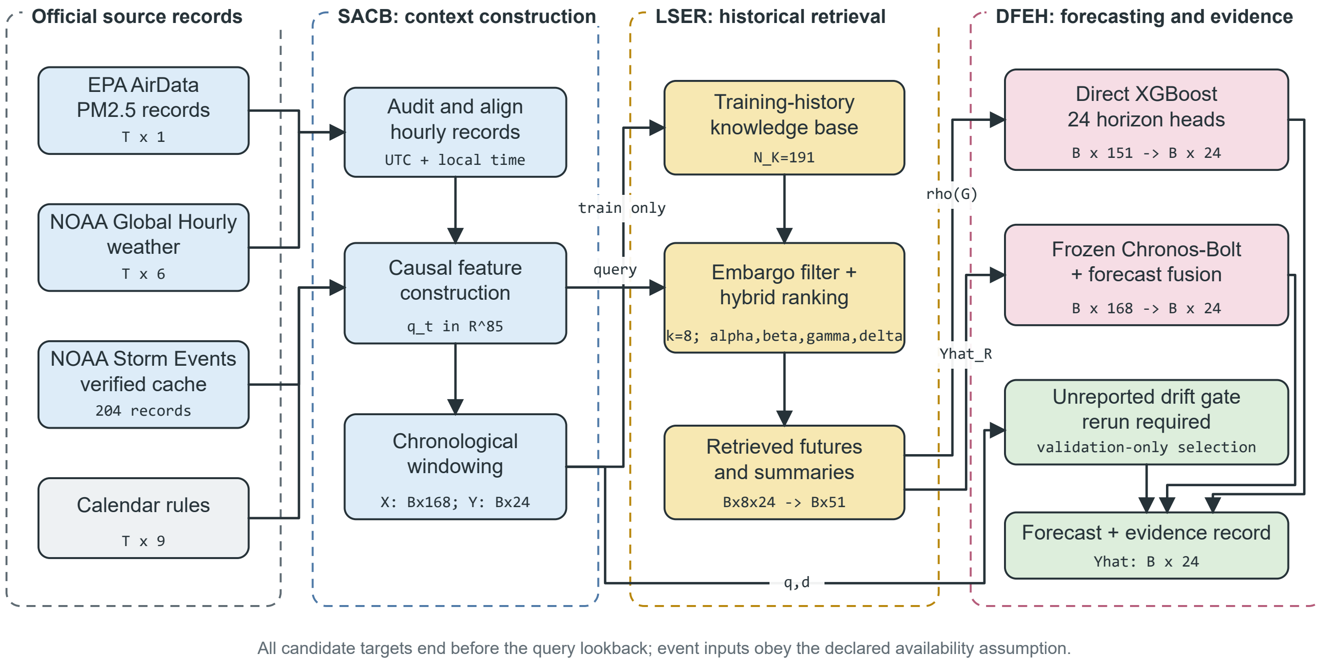

Fig. 1. Proposed Event-TimeRAF pipeline. Official records pass through the Source-Audited Context Builder (SACB), training-history candidates are selected by the Leakage-Safe Event-Context Retriever (LSER), and the Drift-Aware Forecast and Evidence Head (DFEH) produces 24-hour forecasts and traceable evidence. Tensor annotations show the reported configuration.

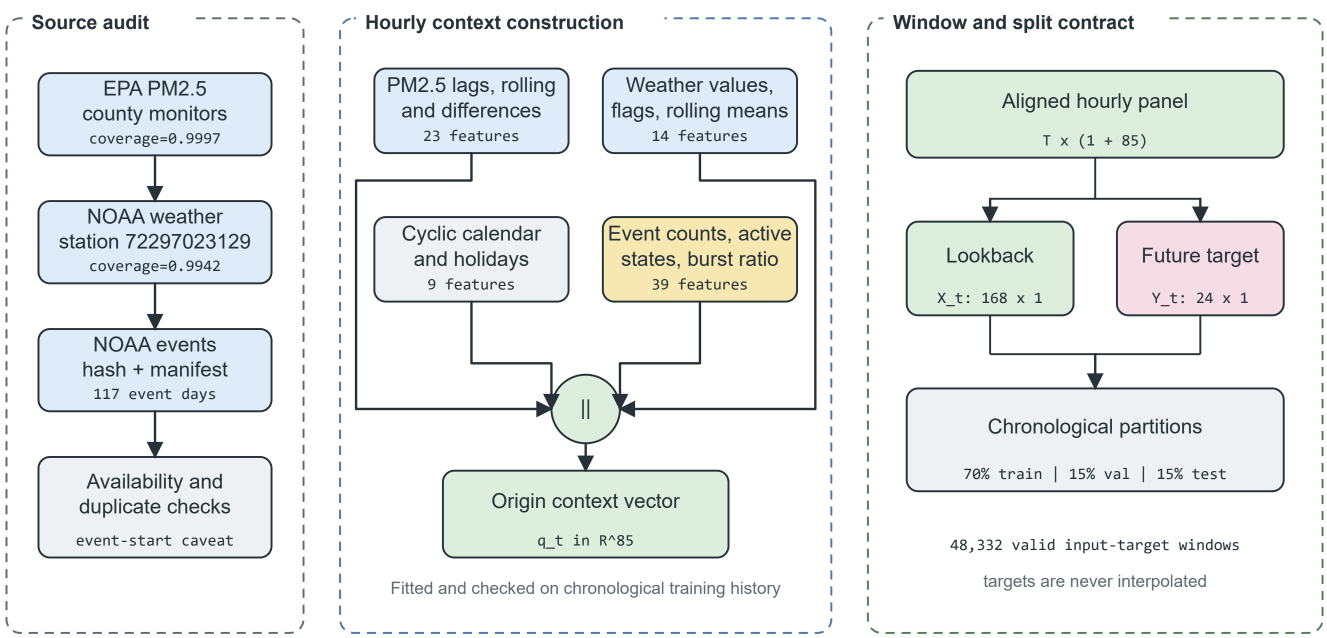

Fig. 2. Source-Audited Context Builder. Source checks feed four feature groups that form an 85-dimensional origin vector. The aligned panel is converted into 168-hour inputs and 24-hour targets before the chronological split.

is a calendar summary, and e_i is the event-context summary available at u_i. For a query origin t, the temporally eligible set is

$$\mathcal{A}_t = \{i : u_i + H < t - L + 1,\ i \in \mathcal{T}_{train}\}, \qquad (9)$$

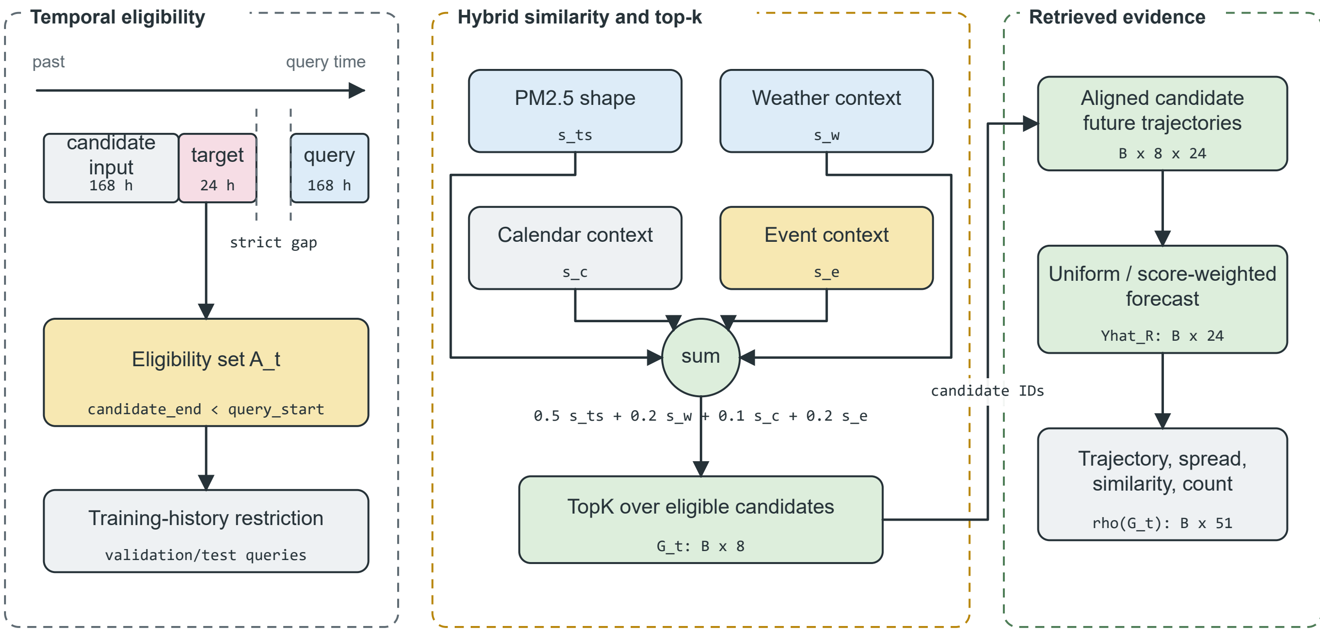

Fig. 3. Leakage-Safe Event-Context Retriever. A strict candidate-target embargo and training-history restriction are applied before hybrid ranking. The top eight candidates provide aligned future trajectories and a 51-dimensional retrieval summary.

for validation and test queries. The strict inequality in (9) ensures that a candidate's target period finishes before the query input period begins. The training-history condition in (9) is also enforced for final validation and test retrieval. Equation (8) defines the stored candidate schema, while Equation (9) defines the admissible subset at query time.

Given a query window X_t, the time-series retriever returns

$$R_t^{ts} = \{r_{t,1}^{ts}, r_{t,2}^{ts}, \ldots, r_{t,k}^{ts}\}, \tag{10}$$

where k is the number of retrieved candidates. The default is $k = 8$, with sensitivity analysis for $k \in \{1, 4, 8, 16\}$. Equation (10) denotes the returned records before their future trajectories are summarized.

2) Event Knowledge Base: The event knowledge base contains source-preserving records relevant to Los Angeles County. NOAA Storm Events is the required structured source, whereas NOAA Hazard Mapping System fire and smoke records remain an unmeasured optional extension [36], [37]. Stored fields include event identifier, start and end, information-availability assumption, location, category, source, source URL, title, and summary. The reported records cover wildfire, high wind, heavy rain, flood, and other weather categories; feature slots for absent configured categories remain zero.

Model inputs represent events with hourly counts, rolling 24-, 72-, and 168-hour counts, category-specific 72-hour counts, active-event counts, and an event-burst ratio. Recent source identifiers are retained as explanation evidence. Target-horizon overlap is used only as a post-hoc subset label and never as a model input. NOAA Storm Events does not provide a machine-readable publication timestamp, so the implementation records event start as a retrospective availability assumption. Strict operational experiments require records with genuine publication or issue times.

3) Hybrid Ranking and Future Alignment: Both query and candidate PM2.5 windows are standardized by their own means and standard deviations and normalized to unit length. Weather and calendar groups are standardized using candidate statistics and mapped cosine similarity to $[0, 1]$. Event similarity is clipped to $[0, 1]$ with explicit zero-vector handling. The hybrid retrieval score combines time-series similarity, weather relevance, calendar relevance, and event relevance:

$$s(q, i) = \alpha s_{ts}(q, i) + \beta s_w(q, i) + \gamma s_c(q, i) + \delta s_e(q, i), \tag{11}$$

where s_{ts} is time-series similarity, s_w is weather similarity, s_c is calendar similarity, and s_e is event relevance. The implemented weights are $\alpha = 0.5$, $\beta = 0.2$, $\gamma = 0.1$, and $\delta = 0.2$. Equation (11) keeps each relevance channel explicit so that its contribution can be inspected without a learned latent scoring function.

The retrieved set is selected as

$$G_t = \mathrm{TopK}_{i \in \mathcal{A}_t} \; s(t, i). \tag{12}$$

Equation (12) is evaluated only after applying $\mathcal{A}_t$. Candidate futures are restored to the query scale as

$$\widetilde{Y}_{t,i} = \mu_t + \sigma_t \left(\frac{Y_i - \mu_i}{\max(\sigma_i, \epsilon)} \right), \qquad i \in G_t, \tag{13}$$

where (μ_t, σ_t) and (μ_i, σ_i) are the query and candidate lookback statistics. Retrieval-only forecasts average $\widetilde{Y}_{t,i}$ uniformly or by non-negative normalized scores. The feature-level path retains the mean trajectory, horizon-wise spread, mean and maximum time-series similarity, and candidate count, yielding $\rho(G_t) \in \mathbb{R}^{51}$. The affine transformation in Equation (13) preserves the candidate trajectory shape while restoring the query's local scale.

D. Drift-Aware Forecast and Evidence Head

The DFEH is the common output layer for two verified forecasting paths and the evidence output. As shown in Figure 4, the direct path concatenates context, retrieval, drift, and future-calendar features for 24 separately trained XGBoost regressors. A second path evaluates frozen Chronos-Bolt and combines its forecast with the retrieved forecast at a validation-selected weight. The current code also defines a validation-only drift-state selector, but no manifest-backed run of that selector is reported in this manuscript.

1) Feature-Level Forecasting and Drift Gate: For horizon h, the direct model input and prediction are

$$u_{t,h} = [q_t \,\|\, \rho(G_t) \,\|\, d_t \,\|\, c_{t+h}], \qquad \hat{y}^{X}_{t+h} = f_h(u_{t,h}), \quad (14)$$

where $u_{t,h} \in \mathbb{R}^{151}$ for the full M09 model and each f_h is an XGBoost regressor trained with squared-error objective. This design gives each forecast horizon a separate supervised function while retaining a common origin representation. Equation (14) also accounts for the verified 151 features: 85 context, 51 retrieval, six drift, and nine future-calendar values.

The tabular fusion stage includes a lightweight drift-conditioned selector. It compares the strongest validation candidates separately on drift and non-drift validation origins, then applies the selected source to the corresponding test origins:

$$\hat{Y}^{G}_t = \begin{cases} \hat{Y}^{a}_t, & D_t = 0, \\ \hat{Y}^{b}_t, & D_t = 1, \end{cases} \quad (15)$$

where a and b are chosen only from validation errors. This gate does not train a new model; it tests whether a transparent drift-specific selection rule can avoid applying the same forecaster in all regimes. Equation (15) describes the current notebook procedure. It is excluded from the reported comparison until a complete rerun records its selection, predictions, metrics, and checksums in one manifest.

2) Frozen-TSFM Forecast Fusion: The Chronos-Bolt retrieval variant uses forecast-level fusion rather than internal Channel Prompting. Let $\hat{Y}^{C}_t$ be the frozen Chronos forecast and $\hat{Y}^{R}_t$ be the retrieved historical forecast. The fused prediction is

$$\hat{Y}^{CR}_t(\omega) = \omega \hat{Y}^{C}_t + (1-\omega)\hat{Y}^{R}_t, \quad (16)$$

where $\omega \in \{0, 0.25, 0.5, 0.75, 1.0\}$ is selected on the validation split. Validation selected $\omega = 0.75$. Because fusion occurs after both forecasts are produced, this branch does not modify Chronos-Bolt embeddings or weights. Equation (16) is therefore an output-space interpolation, not an internal TSFM prompting operation.

3) Concept-Drift Evidence: The implementation represents distribution shift using transparent indicators rather than claiming perfect concept-drift identification. The indicators are recent mean shift, recent variance shift, retrieval similarity drop, weather-pattern shift, and event-burst ratio. The raw vector used by the code is

$$r_t = \Bigg[|\mu_{24,t} - \mu_{168,t}|, \left| \log \frac{\sigma_{24,t} + \epsilon}{\sigma_{168,t} + \epsilon} \right|, 1 - \mathrm{clip}(\bar{s}_{ts,t}, 0, 1), \sqrt{\frac{1}{4}\sum_{\ell=1}^{4} \left(\frac{w_{t,\ell} - \widetilde{w}^{tr}_{\ell}}{s^{tr}_{w,\ell}} \right)^2}, \; \max(b_{event,t}, 0) \Bigg], \quad (17)$$

where the weather term uses temperature, relative humidity, pressure, and wind speed standardized by training medians and standard deviations. Each component $r_{t,j}$ is then robustly scaled using the training median and median absolute deviation (MAD):

$$d_{t,j} = \frac{1}{6}\,\mathrm{clip}\left(\frac{r_{t,j} - \mathrm{med}_{tr}(r_j)}{\max(1.4826\,\mathrm{MAD}_{tr}(r_j), \epsilon)}, 0, 6 \right). \quad (18)$$

Finally, the continuous score and binary flag are

$$S_t = \frac{1}{5}\sum_{j=1}^{5} d_{t,j}, \qquad D_t = \mathbb{I}[S_t \geq Q_{0.90}(\{S_v : v \in \mathcal{T}_{val}\})]. \quad (19)$$

Equations (17)–(19) match the implemented detector: training data set the reference statistics, validation data set the 0.90-quantile threshold, and test data are transformed without refitting. The binary D_t is the state used in Equation (15).

4) Evidence Record and Uncertainty Proxy: The explanation module uses signed local XGBoost contributions averaged across the 24 direct models, weather flags, the top three retrieved windows, recent event records, calendar context, forecast direction, and component-level drift evidence. Its uncertainty proxy is

$$u^{\mathrm{proxy}}_t = \sqrt{\bar{\sigma}^2_{R,t} + \bar{e}^2_{val}}, \quad (20)$$

where $\bar{\sigma}_{R,t}$ is the mean horizon-wise spread of the retrieved trajectories and $\bar{e}_{val}$ is the mean absolute validation residual. The result is a diagnostic scale, not a calibrated prediction interval. Each explanation is generated from stored fields rather than an unrestricted language model, and each event statement remains contextual rather than causal. Equation (20) is retained only as an evidence-field definition and is not used to construct confidence intervals.

E. Relationship to TimeRAF

TimeRAF retrieves top-k time-series windows using a learned MLP dual encoder and integrates candidate embeddings into a frozen TSFM through Channel Prompting [9]. Table II identifies which principles are retained and which mechanisms differ. The comparison is methodological rather than a performance ranking because the studies use different datasets, horizons, and protocols.

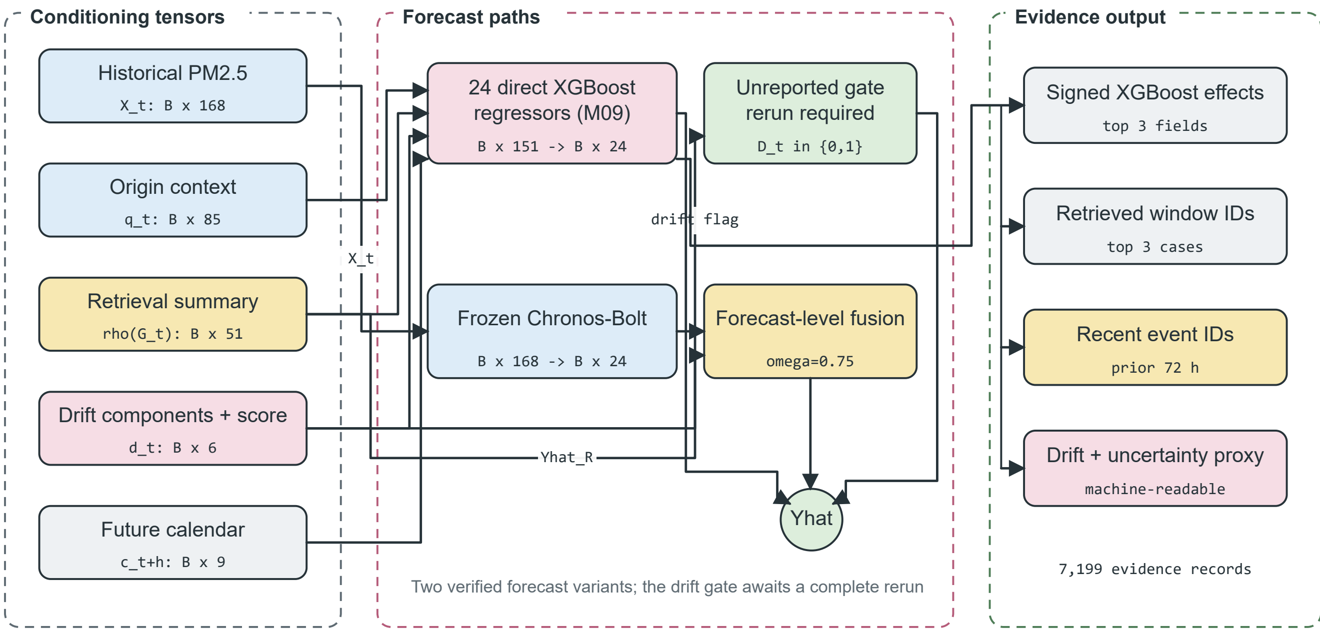

Fig. 4. Drift-Aware Forecast and Evidence Head. Context, retrieval, drift, and future-calendar tensors feed direct horizon regressors; the frozen Chronos-Bolt path is fused only at forecast level. Stored feature effects, retrieved cases, event identifiers, drift components, and uncertainty values form the evidence record.

TABLE II
METHODOLOGICAL COMPARISON OF TIMERAF AND EVENT-TIMERAF

Aspect	TimeRAF	Event-TimeRAF
Primary objective	Improve zero-shot forecasting by augmenting a frozen TSFM with external time-series knowledge.	Evaluate source-audited, event-context retrieval for local PM2.5 forecasting under temporal shift.
Forecasting domain	Multi-domain benchmark datasets.	Los Angeles County hourly PM2.5, 2019–2024.
Knowledge base	Curated, consistently normalized time-series windows.	191 non-overlapping training-history windows with PM2.5, weather, calendar, and retrospective event context.
Retriever	Learned MLP dual encoder with forecaster-guided training.	Random, cosine, and transparent hybrid similarity; no learned retriever.
Knowledge integration	Internal Channel Prompting of candidate embeddings.	Feature-level XGBoost inputs and forecast-level Chronos-Bolt fusion.
Backbone	Frozen TSFM, principally TTM-Base.	Direct XGBoost horizon models and frozen `amazon/chronos-bolt-small`.
Leakage control	Non-overlapping knowledge windows and cross-dataset training constraints.	Candidate target must end before query lookback; validation and test retrieval use training history only.
Event handling	Not an explicit module.	Availability-tagged NOAA Storm Events features and source-record evidence, reported retrospectively.
Drift and explanation	Retrieved-case inspection.	Five-component drift audit, signed feature effects, retrieved IDs, event IDs, and uncertainty proxy.
Implementation scope	Learned retriever and Channel Prompting are implemented.	Those mechanisms are not implemented and remain future extensions.

VII. EXPERIMENTAL SETUP

This section describes the reported experiment. Model-selection choices for M00–M11 were fixed using the training and validation partitions before the test results were examined. A later post-hoc selector is excluded from every reported comparison because it was added after the manifest was written. Artifact consistency was verified using run identifier `20260723T112033170131Z`.

A. Dataset Description and Audit

The study period is 2019–2024. PM2.5 observations were taken from US EPA AirData/AQS parameter 88101 for Los Angeles County [34]. No single monitoring site was used as the target series; the audited target is the hourly county median across official Los Angeles County AQS monitors. This fall-back selected `06-037-COUNTY`, with 52,602 observed target hours and observed PM2.5 coverage of 0.9997. The PM2.5 unit was consistently recorded as micrograms per cubic meter.

Weather covariates were taken from NOAA Global Hourly/ISD through the official NOAA Open Data Dis-

TABLE III
DATA-READINESS AUDIT

Gate	Value	Status
PM2.5 observed coverage	0.9997	Pass
Weather complete coverage	0.9942	Pass
Event days	117	Pass
Event overlap days	2,192	Pass
Qualifying event categories	3	Pass
Strict event availability	False	Caveat

semination public buckets [35]. The selected station was `72297023129`, Long Beach / Daugherty Field / Airport, 10.28 km from the PM2.5 target centroid. The complete weather-feature coverage after allowed past-only short-gap filling was 0.9942, and the raw measured core coverage was 0.9905.

Environmental events were loaded from hash-verified NOAA Storm Events annual detail archives [36]. Because direct access to NCEI was unavailable in the execution environment, the experiment used a locally staged cache containing unchanged NOAA annual archives and a source manifest with official URLs, byte counts, and SHA-256 hashes. The audit retained the delivery mode `verified_official_noaa_cache`. The event audit found 117 event days, 2,192 source-overlap days, and three qualifying event categories. NOAA Storm Events does not expose machine-readable publication timestamps; therefore, event-aware outputs use the recorded event start time as a retrospective availability assumption. They are not strict real-time event forecasts. Table III reports the resulting readiness status. The source audit and window contract are visualized in Figure 2.

Table IV consolidates the evaluated scope and the role of each source in the forecasting pipeline.

B. Data and Code Availability

The raw public records and download pages are available from EPA AirData, NOAA Global Hourly/ISD, and NOAA Storm Events [34]–[36]. The experiment used a locally staged NOAA Storm Events cache because direct NCEI access was unavailable in the execution environment; every cached NOAA file was verified against the generated source manifest. Source code is available at https://github.com/shasabbir/event-time-raf. The implementation uses Python 3.12.13, NumPy, pandas, scikit-learn, XGBoost, PyTorch, and the `chronos-forecasting` package; the repository contains the configuration, notebooks, source modules, and tests needed to reproduce the pipeline when the public source files are accessible.

C. Forecasting Task and Split

The forecasting task predicts PM2.5 from one to 24 hours ahead using a 168-hour lookback window. The split is chronological: the earliest 70% of valid windows are used for training, the next 15% for validation, and the latest 15% for testing. The final test set contains 7,199 forecast origins and 172,776 forecasted hourly target points. Random splitting is not used.

D. Models

The evaluated models are persistence, daily seasonal naive, weekly seasonal naive, a direct XGBoost model using PM2.5 features, a direct XGBoost context model using PM2.5, weather, and calendar features, random retrieval, cosine retrieval, XGBoost with cosine retrieval features, Event-TimeRAF without drift features, full Event-TimeRAF with event and drift features, frozen Chronos-Bolt, and retrieval-augmented Chronos-Bolt forecast fusion. An auxiliary random-retrieval XGBoost model A01 and an event-free model A00 support the ablation analysis. A post-hoc M12 artifact is excluded because it is not part of the manifest-backed result set. The foundation-model evaluation uses the `amazon/chronos-bolt-small` checkpoint. Table V groups the evaluated configurations by their forecasting and retrieval roles.

E. Implementation and Hyperparameters

The experiment was executed in a cloud-hosted Linux 6.12.90 environment using Python 3.12.13. The runtime was 1,576.87 seconds. Key package versions recorded in the run manifest include NumPy 2.0.2, pandas 2.3.3, scikit-learn 1.6.1, XGBoost 3.2.0, PyTorch 2.10.0+cu128, and chronos-forecasting 2.3.1. Table VI reports the principal hyperparameters.

F. Evaluation Metrics and Statistical Checks

The primary metrics are MSE, MAE, RMSE, MAPE, sMAPE, and R^2. MSE is retained for comparability with the TimeRAF evaluation, while MAE and RMSE provide more interpretable PM2.5 error scales. For n observed forecast values y_j and predictions $\hat{y}_j$, the principal error metrics are

$$\begin{aligned} \mathrm{MSE} &= \frac{1}{n}\sum_{j=1}^{n}(y_j-\hat{y}_j)^2, \\ \mathrm{MAE} &= \frac{1}{n}\sum_{j=1}^{n}|y_j-\hat{y}_j|, \\ \mathrm{RMSE} &= \sqrt{\mathrm{MSE}}, \end{aligned} \tag{21}$$

and explained variance is summarized by

$$R^2 = 1-\frac{\sum_{j=1}^{n}(y_j-\hat{y}_j)^2}{\sum_{j=1}^{n}(y_j-\bar{y})^2}. \tag{22}$$

Equation (21) is aggregated across all forecast origins and horizons for the overall result, while (22) uses the same set of points. MAPE and sMAPE are retained in machine-readable outputs but are not used for the main ranking because low PM2.5 values can make percentage errors unstable. Subset metrics are reported only when at least 50 forecast origins are available. The event subset contains 532 origins, the active-event subset contains 293 origins, and the drift subset contains 560 origins, so all pre-specified subsets satisfy the minimum sample-size criterion. Pairwise comparisons use paired block bootstrap intervals with 24-hour blocks and 500 resamples.

TABLE IV
DATASET AND SOURCE SUMMARY

Data group	Source	Evaluated scope	Use in pipeline
PM2.5	EPA AirData/AQS hourly parameter 88101	Los Angeles County, 2019–2024; 52,602 observed target hours	Target sequence and lag/rolling features
Weather	NOAA Global Hourly/ISD, station `72297023129`	2019–2024; 0.9942 usable core coverage	Temperature, humidity, wind, pressure, and rain features
Events	NOAA Storm Events annual detail archives	204 Los Angeles records; 117 event days	Retrospective event-start features and explanation evidence
Calendar	Generated from timestamps and public holiday rules	2019–2024 hourly calendar	Hour, weekday, weekend, month, season, and holiday features

TABLE V
EVALUATED MODEL FAMILIES

ID	Description
M00–M02	Persistence and seasonal naive baselines
M03–M04	XGBoost PM2.5-only and context models
M05–M06	Random and cosine retrieval-only baselines
M07	XGBoost with cosine retrieval features
M08–M09	Event-TimeRAF without/with drift indicators
M10–M11	Frozen Chronos-Bolt and retrieval fusion

G. Comparison with Existing Studies

Direct numerical comparison with existing TSFM and air-quality papers is not valid because they use different datasets, forecast horizons, variables, and evaluation protocols. Table VII therefore compares methodological coverage rather than claiming cross-paper performance dominance.

VIII. RESULTS AND DISCUSSION

A. Overall Comparison with Baselines

Table VIII reports the test-set results. The strongest pre-specified supervised result is M04, which uses PM2.5 history, weather, and calendar features and obtains MSE 26.185, MAE 3.125, RMSE 5.117, and R^2 0.379. The full feature-level Event-TimeRAF model M09 obtains MSE 26.712 and MAE 3.149; event-context retrieval and drift features therefore do not improve the all-origin error over M04 in this run. A selector appended after the archived run is omitted: its result tables and prediction file do not match the run manifest, so it is not part of the verified comparison.

The frozen Chronos-Bolt evaluation completed successfully. Validation selected a retrieval-fusion weight of 0.75. On the test set, retrieval-augmented Chronos reduces MSE from 28.941 to 28.709, an improvement of about 0.8% relative to frozen Chronos-Bolt. The paired block-bootstrap comparison gives an MSE difference of -0.233 with a 95% interval from -0.665 to 0.152, so the direction is favorable but not conclusive under this uncertainty estimate. MAE changes from 3.205 to 3.209, which is practically flat.

B. Ablation Study

Table IX summarizes the component-wise ablation comparisons. The clearest supported gain is from adding weather and calendar context to the PM2.5-only XGBoost model: MSE decreases by 3.764 and the confidence interval is fully below zero. Cosine retrieval improves over random retrieval in the retrieval-only setting. In contrast, adding cosine retrieval features to the already strong XGBoost context model does not improve the all-origin result in this run. Event-hybrid retrieval and drift indicators also do not show a clear overall improvement beyond the context baseline. Comparing M09 against the event-free A00 model also yields an interval that crosses zero, so the event component has no verified overall gain in this run. The auxiliary A00 and A01 models are retained only to isolate event and random-retrieval effects.

The ablation does not form a monotone accuracy progression: weather-calendar context earns its place, cosine retrieval is useful relative to random retrieval in the retrieval-only path, and the measured event and drift additions do not improve the strongest context model. This non-monotonic pattern is retained rather than implying that every named component increases accuracy.

The k sensitivity results in Table X show that the retrieval-only cosine baseline improves as more neighbors are used within the tested range. The best retrieval-only configuration among $k \in \{1, 4, 8, 16\}$ is $k = 16$, with MSE 36.488. The reported Event-TimeRAF feature models used the pre-specified default $k = 8$.

C. Event and Drift Subsets

Table XI reports representative event and drift subsets for the strongest context model, the full Event-TimeRAF model, and the two Chronos variants. All subsets satisfy the pre-specified minimum sample-size criterion. The event subset has lower MSE than the non-event subset for these models, which should not be interpreted as events being easier in general; it reflects the particular 2024 test-period event hours retained by the audit. The drift subset is defined by the validation-calibrated DFEH flag and is used only for stratified evaluation.

D. Horizon-Wise and Retrieval Diagnostics

Fig. 5 and Fig. 6 show horizon-wise test errors. Fig. 7 summarizes retrieval behavior, and Fig. 8 shows the drift-score distribution across the test period. Figure 9 shows a representative test-origin forecast, allowing M04 and M09 to be compared with the observed target. These plots are regenerated from predictions identified by `20260723T112033170131Z` and include only M00–M11.

TABLE VI
EXPERIMENTAL HYPERPARAMETERS AND RUNTIME SETTINGS

Component	Setting
Study period	2019–2024
Lookback and horizon	$L = 168$ hours, $H = 24$ hours
Split	70% train, 15% validation, 15% test, chronological
SACB feature groups	23 PM2.5, 14 weather, 9 calendar, 39 event features
SACB missing-data rule	Past-only filling for gaps up to 3 hours; no target interpolation
LSER retrieval	$k = 8$, sensitivity for $k \in \{1, 4, 8, 16\}$
LSER knowledge base	191 windows; 192-hour stride; training history only
LSER similarity weights	time series 0.5, weather 0.2, calendar 0.1, event 0.2
DFEH direct input	151 features for each of 24 horizon regressors
DFEH XGBoost	350 estimators, max depth 6, learning rate 0.05, subsample 0.85, colsample 0.85, $\lambda = 1.0$
DFEH drift detector	5 components; 24-hour recent window, 168-hour reference window, 0.90 validation quantile threshold
DFEH Chronos path	`amazon/chronos-bolt-small`, batch size 64
DFEH fusion weights	$\{0, 0.25, 0.5, 0.75, 1.0\}$; selected $\omega = 0.75$
Explanation record	Top 3 feature effects, windows, and recent event IDs
Bootstrap	500 paired block-bootstrap resamples, 24-hour blocks

TABLE VII
METHODOLOGICAL COMPARISON WITH RELATED FORECASTING WORK

Method family	TSFM	Retrieval	Events	Explainability	Comment
PatchTST-style Transformers [22]	No	No	No	Limited	Strong sequence modeling, but event context is not central.
Chronos [5]	Yes	No	No	Limited	Frozen zero-shot baseline used in this study.
TimeRAF [9]	Yes	Yes	No	Retrieval cases	Base retrieval-augmented TSFM method; does not target official event-aware PM2.5.
XGBoost context baseline [10]	No	No	No	Feature importance	Strong supervised tabular baseline for local PM2.5.
Event-TimeRAF	Yes	Yes	Yes	Evidence-grounded	Adds official event audit, temporal leakage control, drift indicators, and Chronos retrieval fusion.

TABLE VIII
MANIFEST-BACKED OVERALL TEST RESULTS FOR 24-HOUR PM2.5 FORECASTING

Model	MSE	MAE	RMSE	R^2
M00 persistence	47.649	4.137	6.903	-0.130
M01 daily seasonal	48.706	4.057	6.979	-0.155
M02 weekly seasonal	65.606	5.123	8.100	-0.556
M03 XGBoost PM2.5	29.949	3.368	5.473	0.290
M04 XGBoost context	**26.185**	**3.125**	**5.117**	**0.379**
M05 random retrieval	42.367	4.257	6.509	-0.005
M06 cosine retrieval	38.083	3.940	6.171	0.097
M07 XGBoost cosine	26.672	3.151	5.164	0.367
M08 Event-TimeRAF no drift	26.677	3.145	5.165	0.367
M09 Event-TimeRAF full	26.712	3.149	5.168	0.367
M10 frozen Chronos-Bolt	28.941	3.205	5.380	0.314
M11 Chronos + retrieval	28.709	3.209	5.358	0.319

TABLE IX
SELECTED PAIRED BLOCK-BOOTSTRAP ABLATION RESULTS

Comparison	MSE diff.	95% CI low	95% CI high
M04 minus M03, SACB weather/calendar	-3.764	-5.856	-2.071
M06 minus M05, LSER cosine/random ranking	-4.284	-5.862	-2.759
M07 minus M04, LSER retrieval summary	0.487	0.015	1.028
M08 minus M07, LSER event-hybrid ranking	0.005	-0.292	0.275
M09 minus M08, DFEH drift indicators	0.035	-0.101	0.161
M09 minus A00, event context	0.083	-0.225	0.354
M09 minus M04, SACB+LSER+DFEH	0.528	-0.026	1.182
M11 minus M10, DFEH Chronos fusion	-0.233	-0.665	0.152

E. Explainability and Evidence Analysis

Within the DFEH, explainability is implemented as a source-grounded evidence record rather than as a post-

TABLE X
COSINE RETRIEVAL SENSITIVITY TO k

k	MSE	MAE	RMSE
1	65.129	5.133	8.070
4	41.237	4.134	6.422
8	38.083	3.940	6.171
16	36.488	3.851	6.040

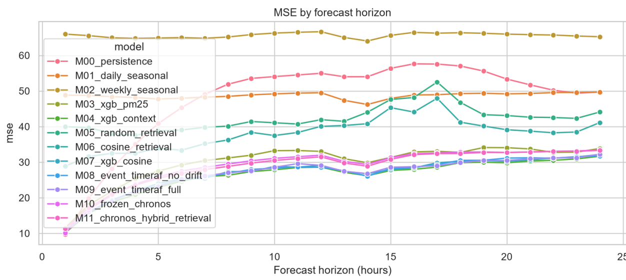

Fig. 5. MSE by forecast horizon on the Los Angeles County test set.

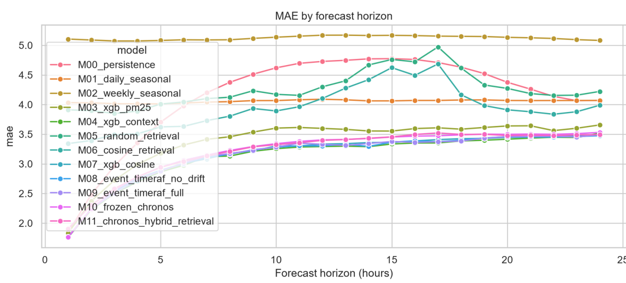

Fig. 6. MAE by forecast horizon on the Los Angeles County test set.

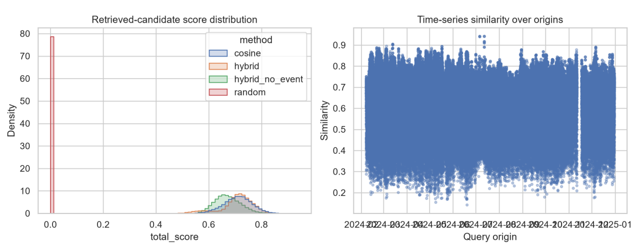

Fig. 7. Similarity and candidate diagnostics for test-set retrieval.

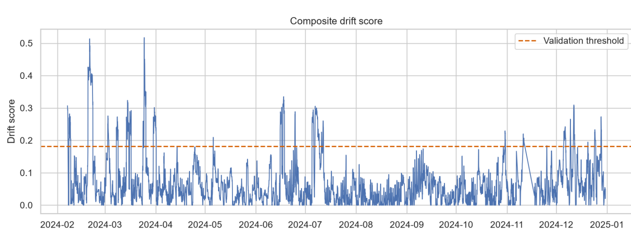

Fig. 8. Composite drift-score diagnostics on the test set.

hoc free-text interpretation. For each evaluated forecast origin, the pipeline stores the run identifier, window identifier, origin timestamp, retrieved historical evidence identifiers, event evidence identifiers, leading feature effects, drift components, retrieval spread, validation residual MAE, uncertainty proxy, drift score, drift flag, and event-availability mode. The evaluation produced 7,199 explanation records in `explanations.parquet`. A representative record identifies the strongest local contribution fields as the current PM2.5 value and recent rolling PM2.5 means, links the prediction to three retrieved historical windows, links recent source-recorded storm events when available, and marks distribution shift when the drift detector is active.

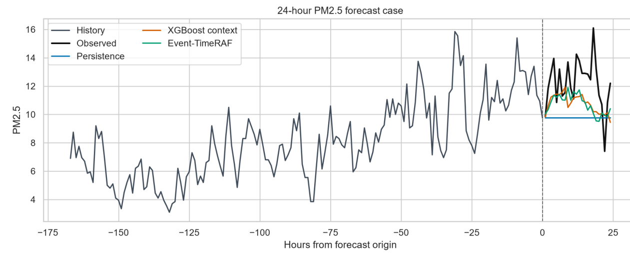

Fig. 9. Representative 24-hour forecast case from the test set.

Table XII summarizes the explanation layer. This design supports auditability because every explanatory statement can be traced back to a model feature, a retrieved time-series window, an official event record, or a drift component. It does not prove causality: event evidence is reported as contextual support, and the drift flag is treated as an uncertainty warning.

F. Computational Cost and Reproducibility

The complete experiment executed in 1,576.9 seconds in a cloud-hosted Linux environment using Python 3.12.13. The recorded package versions include NumPy 2.0.2, pandas 2.3.3, scikit-learn 1.6.1, XGBoost 3.2.0, PyTorch 2.10.0+cu128, and `chronos-forecasting` 2.3.1. The run manifest records the configuration SHA-256 hash and artifact checksums. The CPU, GPU, and RAM model identifiers were not saved in the manifest; therefore this paper reports runtime and software versions but does not claim hardware-normalized efficiency. Table XIII lists the reproducibility fields retained by the archived run.

The main interpretation is that conventional supervised context features are very strong for this Los Angeles County 24-hour PM2.5 task. Retrieval is useful as a standalone historical-pattern baseline and gives a small favorable MSE change when fused with frozen Chronos-Bolt. However, the present feature-level Event-TimeRAF design does not beat the best XGBoost context baseline overall. Event-TimeRAF should therefore be presented as an implemented and audited event-aware retrieval framework with selective gains, not as a uniformly superior forecasting model. Post-manifest artifacts are excluded from this interpretation.

Finally, all event-aware interpretations must be read with the availability caveat. NOAA Storm Events records are official and hash-verified, but the source does not provide machine-readable publication timestamps. The experiment records event start as the availability time, so event-related results are retrospective sensitivity results. A strict operational study would require event or smoke sources with genuine issue or

TABLE XI
REPRESENTATIVE EVENT AND DRIFT SUBSET RESULTS

Subset	Model	Origins	MSE	MAE
Non-event	M04 XGBoost context	6,667	27.520	3.179
Event	M04 XGBoost context	532	9.451	2.450
Non-event	M09 Event-TimeRAF full	6,667	27.942	3.187
Event	M09 Event-TimeRAF full	532	11.301	2.666
Non-drift	M04 XGBoost context	6,639	26.107	3.088
Drift	M04 XGBoost context	560	27.110	3.562
Non-drift	M09 Event-TimeRAF full	6,639	26.330	3.088
Drift	M09 Event-TimeRAF full	560	31.244	3.872
Drift	M10 frozen Chronos-Bolt	560	28.333	3.476
Drift	M11 Chronos + retrieval	560	28.198	3.580

TABLE XII
EXPLANATION EVIDENCE STORED BY EVENT-TIMERAF

Evidence field	Interpretation role
Top feature effects	Local model drivers from the direct forecaster.
Retrieved windows	Historical analogues used by the retrieval module.
Event evidence IDs	Official storm-event context linked to the origin.
Drift components	Mean, variance, weather, similarity, and event-burst shift signals.
Uncertainty proxy	Combined retrieval spread and validation residual error.

TABLE XIII
COMPUTATIONAL AND REPRODUCIBILITY SUMMARY

Item	Recorded value
Run identifier	`20260723T112033170131Z`
Runtime	1,576.9 seconds
Platform	Cloud-hosted Linux environment
Python	3.12.13
Forecast horizon	24 hours
Lookback window	168 hours
Bootstrap resamples	500
Chronos checkpoint	`amazon/chronos-bolt-small`
Code availability	https://github.com/shasabbir/event-time-raf

publication times, such as an audited NOAA HMS cache or alert product archive.

IX. LIMITATIONS

The first limitation is event availability. NOAA Storm Events records are official and hash-verified, but the public detail files do not provide machine-readable publication timestamps. The SACB therefore uses event start as an availability proxy, making event-conditioned results retrospective sensitivity analyses rather than operational forecasts.

The second limitation is geographic validation. The experiment covers one county-level PM2.5 aggregate from 2019 through 2024. It does not establish transfer to another county, monitoring topology, climate, or emissions regime, and therefore does not satisfy the external-validation standard for a mature generalization claim.

The third limitation concerns model integration. The LSER uses transparent similarity functions rather than a learned retriever, and the DFEH combines Chronos-Bolt and retrieval only at forecast level. Event-TimeRAF therefore does not reproduce TimeRAF's learned dual encoder or internal Channel Prompting.

The fourth limitation concerns artifact lineage. A drift-gated selector was added after the archived run, and the edited prediction and table files no longer match the original manifest checksums. That selector is therefore excluded from the reported results. Hardware model identifiers and peak memory were also absent from the run manifest, so runtime cannot be interpreted as a hardware-normalized efficiency result.

The fifth limitation concerns cross-version replay of the random-retrieval baseline. Repeating the seeded selection under NumPy 2.4.1 does not reproduce the candidates generated under NumPy 2.0.2. The verification notebook therefore reconstructs M05 and A01 from the original, checksum-verified retrieval-evidence rows. This preserves the reported run but means that exact random-baseline retraining requires the archived software environment.

X. FUTURE WORK

The immediate priority is a clean rerun that records any validation-selected drift gate and all downstream artifacts in one manifest. Operational event studies should then replace event-start proxies with archives that expose genuine issue or publication times. External validation should apply the unchanged protocol to a second US county with distinct meteorology and emissions. Longer-term work may evaluate a learned event-context retriever, richer event embeddings, calibrated probabilistic uncertainty, and internal TSFM conditioning. Those extensions should be introduced only with component-wise ablations and without changing the source audit or temporal embargo.

XI. CONCLUSION

This study presented Event-TimeRAF for 24-hour Los Angeles County PM2.5 forecasting. Its Source-Audited Context Builder aligns official environmental records, its Leakage-Safe Event-Context Retriever enforces a strict historical embargo, and its Drift-Aware Forecast and Evidence Head compares direct and frozen-TSFM paths while retaining traceable evidence. On the 2019–2024 test period, the strongest prespecified supervised context model obtains MSE 26.185, MAE 3.125, RMSE 5.117, and R^2 0.379; the full feature-level Event-TimeRAF model obtains MSE 26.712. Retrieval changes frozen Chronos-Bolt MSE from 28.941 to 28.709, but the paired interval crosses zero. The study therefore supports source-audited retrieval as a transparent analysis framework with selective gains, not as a uniformly superior forecaster. Strict event issue times, external geographic validation, and deeper retrieval integration remain necessary before operational or general cross-domain claims are made.

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